C. You Soared Afar With Grace

time limit per test 2 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1400 | Tags: constructive algorithms, data structures, greedy, implementation |
https://codeforces.com/problemset/problem/2084/C

You are given a permutation a and b of length n[□]. You can perform the following operation at most n times:

- Choose two indices i and j ($1 \le i, j \le n$, $i \ne j$), swap a_i with a_j, swap b_i with b_j.

Determine whether a and b can be reverses of each other after operations. In other words, for each $i = 1, 2, \ldots, n$, $a_i = b_{n+1-i}$.

If it is possible, output any valid sequence of operations. Otherwise, output -1.

[□]A permutation of length n is an array consisting of n distinct integers from 1 to n in arbitrary order. For example, [2,3,1,5,4] is a permutation, but [1,2,2] is not a permutation (2 appears twice in the array), and [1,3,4] is also not a permutation (n=3 but there is 4 in the array).

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 10^4$). The description of the test cases follows.

The first line of each test case contains a single integer n ($2 \le n \le 2 \cdot 10^5$) — the length of the permutations.

The second line contains n integers $a_1, a_2, \ldots, a_n$ ($1 \le a_i \le n$).

The third line contains n integers $b_1, b_2, \ldots, b_n$ ($1 \le b_i \le n$).

It is guaranteed that a and b are permutations of length n.

It is guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, if it is impossible, output -1 in the only line. Otherwise, output a single integer m ($0 \le m \le n$) — the number of operations in the first line. In the following m lines, output two integers — the indices i and j ($1 \le i, j \le n$, $i \ne j$) in each operation in order. If there are multiple solutions, print any of them.

Example

Input

```
5
2
1 2
1 2
2
1 2
2 1
4
1 3 2 4
2 4 1 3
5
2 5 1 3 4
3 5 4 2 1
5
3 1 2 4 5
1 2 3 4 5
```

Output

```
-1
0
1
1 2
2
1 2
1 3
-1
```

Note

In the second test case, b is already the reverse of a.

In the third test case, after performing the following operation, b will become the reverse of a:

- Swap a_1, a_2 and swap b_1, b_2. Now a = [3, 1, 2, 4] and b = [4, 2, 1, 3].

In the fourth test case, after performing the following operations in order, b will become the reverse of a:

- Swap a_1, a_2 and swap b_1, b_2. Now a = [5, 2, 1, 3, 4] and b = [5, 3, 4, 2, 1].
- Swap a_1, a_3 and swap b_1, b_3. Now a = [1, 2, 5, 3, 4] and b = [4, 3, 5, 2, 1].